Enhancing Video-LLM Reasoning via Agent-of-Thoughts Distillation

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Abstract

Video Large Language Models (Video-LLMs) have achieved strong performance on video question answering benchmarks, yet they remain limited in their ability to perform explicit multi-step spatial-temporal reasoning and lack interpretability in their inference process. Agent-based systems address these shortcomings by decomposing queries into sub-tasks solved by specialized tools, but they incur substantial computational overhead at inference time. In this work, we propose Agent-of-Thoughts Distillation (AoTD), a framework that bridges this gap by distilling reliable, tool-grounded chain-of-thought (CoT) rationales generated by an agent system into a compact Video-LLM through instruction tuning. Our agent system decomposes video queries into executable programs, invokes specialized models for object detection, temporal grounding, action recognition, and question answering, and converts the resulting execution traces into natural-language CoTs. An LLM-based verifier filters out incorrect or incoherent rationales, yielding 32.3K high-quality CoTs across six datasets. Distilling these CoTs into LLaVA-NeXT-Video-7B yields consistent improvements on both multiple-choice and open-ended video QA benchmarks, while being 4.5× faster and requiring 3.6× less memory than the agent system. Furthermore, we demonstrate that AoTD transfers effectively to other Video-LLM architectures, confirming its generality.

1. Introduction

Video question answering (VideoQA) has emerged as a central benchmark for evaluating multimodal reasoning over dynamic visual content. Recent Video Large Language Models (Video-LLMs) [3, 4, 20, 23, 51] have demonstrated impressive capabilities by leveraging the strong language understanding of pretrained LLMs alongside visual encoders. These models achieve competitive accuracy on established benchmarks through end-to-end training on large-scale video-text data. However, their reasoning process remains largely opaque: given a complex query requiring multi-step spatial or temporal reasoning, a Video-LLM produces an answer without exposing the intermediate steps that led to it, limiting both interpretability and diagnosability.

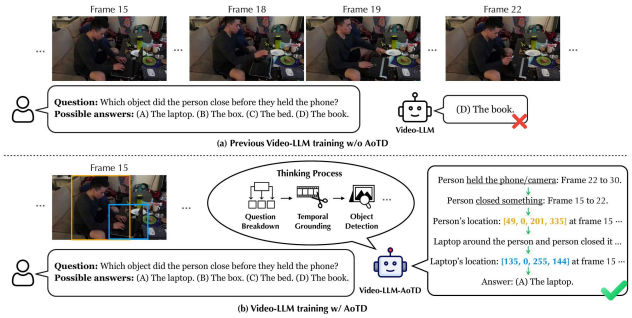

Figure 1. Overview of Agent-of-Thoughts Distillation. An agent system generates multi-step reasoning traces using specialized tools, which are distilled into a Video-LLM that can produce both answers and interpretable rationales at inference time.

Agent-based systems [7, 11, 15, 37] offer an alternative paradigm. By decomposing a user query into a structured program and dispatching sub-tasks to specialized tools—object detectors, temporal grounders, action recognizers—these systems produce transparent execution traces that reveal how an answer was derived. The interpretability comes at a steep cost: each query requires multiple model invocations, leading to high latency and memory consumption that make agent systems impractical for real-time or resource-constrained deployment.

We propose **Agent-of-Thoughts Distillation (AoTD)**, a framework that combines the strengths of both paradigms. AoTD uses an agent system to generate detailed chain-of-thought (CoT) rationales grounded in tool execution traces, verifies these rationales with an LLM-based filter, and then distills them into a Video-LLM via instruction tuning. The resulting model learns to produce both direct answers and step-by-step reasoning without requiring any external tools at inference time, achieving the interpretability of agent systems with the efficiency of end-to-end models.

Our main contributions are as follows:

- We propose AoTD, a novel framework that distills agent-generated, tool-grounded CoT rationales into Video-LLMs, enabling interpretable multi-step reasoning without external tool dependencies at inference time.
- We build an agent system with best-in-class specialized models for each sub-task and introduce an LLM-based verification stage that filters CoTs for correctness and log-

ical coherence.

- We curate a dataset of 32.3K high-quality CoT rationales across six VideoQA datasets covering compositional, temporal, causal, and spatial reasoning.
- We demonstrate consistent improvements on both multiple-choice and open-ended VideoQA benchmarks, 4.5× inference speedup over the agent system, and successful transferability to multiple Video-LLM architectures.

2. Related Work

2.1. Video Large Language Models

The success of Large Language Models has spurred the development of Video-LLMs that extend visual understanding to the temporal domain. Early approaches such as Flamingo [1] and BLIP-2 [19] introduced cross-modal alignment between visual encoders and LLMs. Subsequent work adapted image-based LLMs to video through temporal pooling or frame sampling strategies: LLaMA-VID [22] compresses long videos into compact token representations, Video-LLaVA [23] extends the LLaVA [25] framework with temporal alignment, and VideoChat2 [20] introduces progressive training stages for video conversation. More recent models such as LLaVA-NeXT-Video [51], VideoLLaMA2 [4], and LLaVA-OneVision [3] push performance further through improved training recipes and higher-resolution frame encoding. While these models excel at surface-level recognition, they lack explicit mechanisms for multi-step reasoning over spatial and temporal structures in video.

2.2. Chain-of-Thought Reasoning in VLMs

Chain-of-thought prompting [39] has proven effective for eliciting step-by-step reasoning in language models. Extensions to the visual domain include ViSCoT [36], which generates visual spatial CoTs, CCoT [31], which composes reasoning chains for compositional questions, and multimodal CoT [52], which fuses visual and textual reasoning steps. Tree-of-thoughts [45] further generalizes sequential CoT to branching exploration. Orca [30] demonstrates that distilling step-by-step reasoning from stronger models can substantially improve smaller models. Our work extends this line by generating CoTs not from a single monolithic model but from structured agent execution traces, yielding rationales that are grounded in concrete tool outputs rather than model hallucinations. Process reward models [**?**] provide an orthogonal direction for supervising intermediate reasoning steps, and their integration with tool-grounded rationale generation represents a promising avenue for future work.

2.3. Agent-Based Systems for Video Understanding

Program-based approaches decompose visual queries into executable code. ViperGPT [37] and VisProg [15] synthesize Python programs that invoke vision APIs for image understanding. VideoAgent [7] extends this to video with iterative frame retrieval. AVIS [17] introduces a planner-tool-verifier loop, while DoraemonGPT [44] leverages task-oriented decomposition. ProViQ [5], VideoThought [8], and Cantor [11] further refine program generation and tool orchestration for video reasoning. Additional systems include MoreVQA [28], VideoSTAR [53], VPD [16], and FACT [10]. While these systems achieve strong reasoning performance, their reliance on multiple serial model calls results in high latency and resource consumption. AoTD addresses this limitation by transferring the reasoning capabilities of agent systems into efficient end-to-end models.

2.4. Knowledge Distillation for Vision-Language Models

Knowledge distillation transfers capabilities from larger teacher models to smaller student models. In the vision-language domain, Orca [30] demonstrates that imitating step-by-step explanations from stronger models yields substantial gains at smaller scales, and InstructBLIP [**?**] aligns compact models with GPT-4-level responses through instruction tuning. Unlike these methods that distill from a monolithic model, AoTD distills from a modular agent system whose outputs are grounded in verifiable tool executions, yielding reasoning traces with lower hallucination risk than those from purely generative teachers.

3. Agent-of-Thoughts Distillation

We present Agent-of-Thoughts Distillation (AoTD), a three-stage framework illustrated in Fig. 2: (1) an agent-based system generates chain-of-thought rationales grounded in tool execution traces, (2) an LLM-based verifier filters out incorrect or incoherent rationales, and (3) the verified CoTs are distilled into a Video-LLM through instruction tuning with a dual-prompt strategy.

3.1. Agent-Based System for CoT Generation

Our agent system takes a video and a natural-language query as input and produces both an answer and a structured execution trace. The system operates in two phases: query decomposition and program execution.

Query Decomposition. Given a query q about a video v, a code-generation LLM decomposes q into an executable program p consisting of a sequence of API calls to specialized tools. We use DeepSeek-Coder-Instruct (6.7B) [6] for program generation, which we select based on its superior accuracy (85.7%) compared to alternatives such as

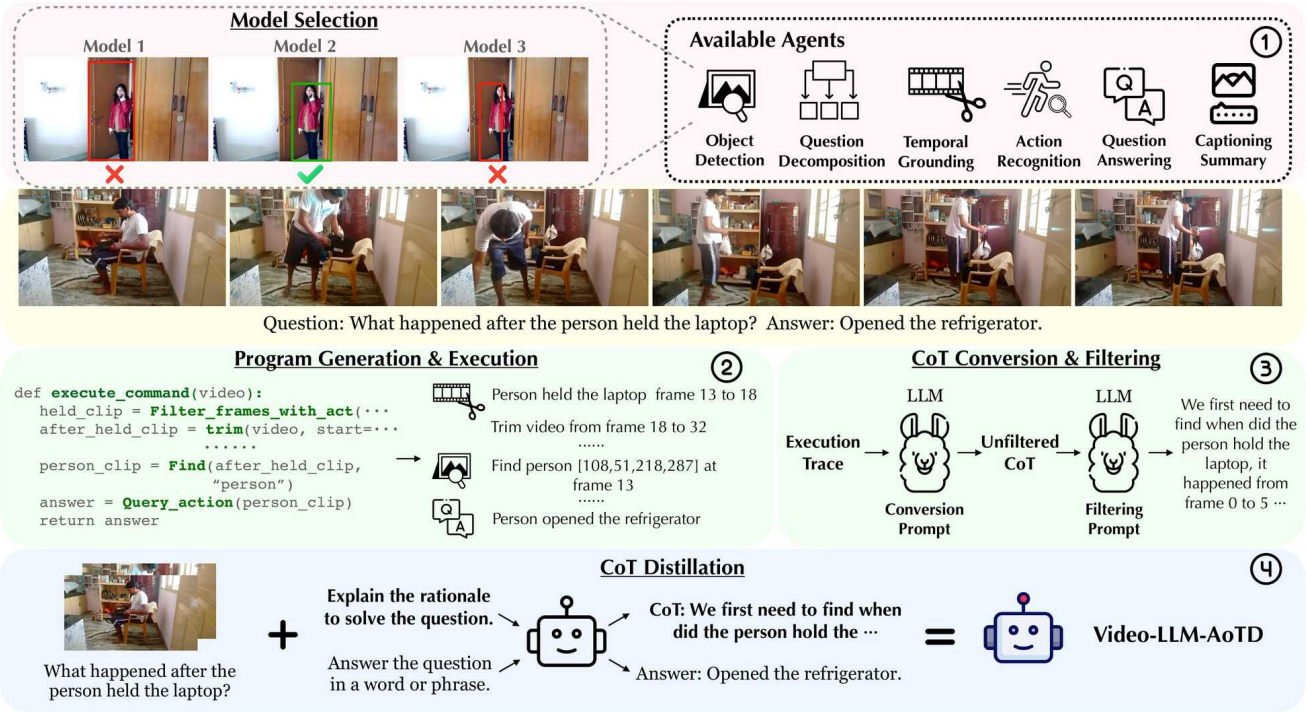

Figure 2. Overview of the AoTD pipeline. Given a video and query, the agent system decomposes the query into an executable program, invokes specialized tools to produce execution traces, and converts these traces into natural-language CoTs. After LLM-based verification, the filtered CoTs are used to instruction-tune a Video-LLM with dual prompts for answer generation and rationale generation.

CodeQwen1.5-Chat [2] and GPT-3.5-Turbo [32] (see Table 1).

Program Execution. The generated program invokes specialized models for each sub-task, as shown in Fig. 3. We evaluate multiple candidate models per sub-task and select the best-performing one:

- **Object Detection**: OWL-ViT v2 [29] (63.0% IoU), selected over OWL-ViT v1 [27] and GLIP [21].
- **Temporal Grounding**: UniVTG [24] (24.7% IoU / 35.3% Recall), selected over LITA [18] and TimeChat [35].
- **Action Recognition**: LLaVA-NeXT-Video-DPO (7B) [51] (18.2% Top1-Acc), selected over InternVideo2 [38] and Open-VCLIP [40].
- **Question Answering**: LLaVA-NeXT-Video-DPO (7B) [51] (53.4% Acc), selected over LLaMA-VID [22] and SeViLA [47].

Trace-to-CoT Conversion. The program execution yields a structured trace containing the program code, intermediate tool outputs, and the final answer. We convert this trace into a natural-language CoT rationale that describes the reasoning steps in a human-readable format, grounding each reasoning step in the corresponding tool output. Each rationale follows a four-part template: (1) a restatement of the query intent, (2) a description of which tools were invoked and the outputs they produced, (3) a synthesis step that integrates intermediate results, and (4) the final answer with an explicit justification. This consistent structure guides the Video-LLM to internalize a reusable reasoning schema during distillation rather

Sub-task	Model name	Metric	Number (%)
question decomposition	CodeQwen1.5-Chat (7B) [2]	Acc	52.7
	GPT-3.5-Turbo [32]		73.1
	DeepSeek-Coder-Instruct (6.7B) [6]		85.7
object detection	OWL-ViT v1 [27]	IoU	47.3
	GLIP [21]		58.9
	OWL-ViT v2 [29]		63.0
temporal grounding	LITA (13B) [18]	IoU / Recall	11.7 / 20.2
	TimeChat (7B) [35]		13.9 / 23.1
	UniVTG [24]		24.7 / 35.3
action recognition	InternVideo2 (1B) [38]	Top1-Acc	7.6
	Open-VCLIP [40]		8.9
	LLaVA-NeXT-Video-DPO (7B) [51]		18.2
question answering	LLaMA-VID (7B) [22]	Acc	43.5
	SeViLA [47]		46.5
	LLaVA-NeXT-Video-DPO (7B) [51]		53.4

Table 1. Sub-tasks definition and evaluation results. We choose 3 model candidates for each sub-task and evaluate them in STAR training set with the corresponding metrics. Models with best performance are placed at the bottom of each column.

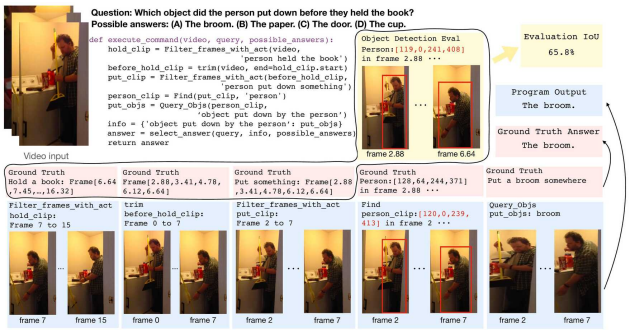

Figure 3. Program execution process in the agent-based system. The decomposed program calls specialized tools sequentially, producing intermediate outputs that form the execution trace.

than memorising surface-level patterns from individual examples.

3.2. CoT Verification and Filtering

Not all agent-generated CoTs are reliable: tool errors, incorrect intermediate results, or flawed program logic can produce misleading rationales. To ensure training data quality, we employ an LLM-based verifier that checks each CoT for factual correctness and logical coherence.

Specifically, we use LLaMA-3.1-8B-Instruct as the verification model. Given a video query, the ground-truth answer, and the generated CoT, the verifier outputs a binary judgment (True/False) indicating whether the rationale correctly supports the answer through logically sound reasoning steps. CoTs judged as incorrect or incoherent are discarded. This filtering stage is critical for distillation quality, as we demonstrate in our ablation study (Sec. 4.4).

3.3. Distillation into Video-LLM

We distill the verified CoTs into a Video-LLM using a dual-prompt instruction tuning strategy. For each training example with an associated CoT, we construct two training instances:

- **Answer-only prompt**: The model is prompted to directly produce the answer given the video and query.
- **Rationale prompt**: The model is prompted to first generate a step-by-step CoT rationale and then produce the answer.

For examples without CoTs, only the answer-only prompt is used. The training objective combines both losses:

$$\mathcal{L} = \mathcal{L}_{\text{answer}} + \mathcal{L}_{\text{rationale}} \tag{1}$$

where both terms are standard next-token prediction losses. This dual-prompt strategy ensures the model retains its direct answering capability while also learning to generate interpretable reasoning chains.

We use LLaVA-NeXT-Video-7B [51] as the base model and fine-tune it with a learning rate of 4×10^{-5}, cosine learning rate schedule, warmup ratio of 0.03, bf16 precision, 32 input frames, and a maximum sequence length of 8192 tokens. Training runs for 3 epochs on 8 A100 80GB GPUs with a global batch size of 128, using gradient checkpointing to reduce memory consumption, and completes in approximately 14 hours. For multi-task mixing, we sample training examples proportionally to dataset size while capping each dataset at 10K examples to prevent domain imbalance.

4. Experiments

4.1. Experimental Setup

Datasets. We construct our training set from six VideoQA datasets spanning compositional, temporal, causal, spatial, and ego-centric reasoning, as summarized in Table 2. Our agent system generates CoTs for four of these datasets, yielding 32.3K verified rationales (Table 3).

Dataset	Size train	Size eval	Type	Train	Eval
MC-VQA					
STAR [41]	45.7K	7.1K	Compositional	✓	✓
NExT-QA [42]	34.1K	5.0K	Temporal & Causal	✓	✓
CLEVRER [46]	21.0K	-	Spatial-temporal	✓	✗
Perception-Test [34]	-	11.5K	General	✗	✓
MVBench [20]	-	4.0K	General	✗	✓
VideoMME [9]	-	2.7K	General	✗	✓
VSIBench [43]	-	5.0K	spatial-temporal	✗	✓
OE-VQA					
AGQA [14]	25.0K	2.0K	Compositional	✓	✓
ANetQA [49]	25.0K	2.0K	Compositional	✓	✓
EgoQA [13]	7.8K	-	Ego-centric	✓	✗
Activitynet-QA [48]	-	8.0K	General	✗	✓
Video-ChatGPT [26]	-	3.0K	General	✗	✓

Table 2. Training and evaluation datasets statics.

Baselines. We compare against proprietary models (GPT-4V [33], Gemini 1.5 Pro [12]) and open-source Video-LLMs including LLaMA-VID [22], Video-LLaVA [23], VideoChat2 [20], VideoLLaMA2 [4], and LLaVA-NeXT-Video [51]. Our primary comparison is between LLaVA-NeXT-Video-AoTD (our model) and LLaVA-NeXT-Video-Instruct (the same architecture fine-tuned with the same data but without CoT rationales).

Evaluation. We evaluate on six multiple-choice (MC-VQA) benchmarks: MVBench [20], VideoMME [9], STAR [41], NExT-QA [42], Perception-Test [34], and VSI-Bench [43]; and four open-ended (OE-VQA) benchmarks: ANetQA [49], AGQA [14], ActivityNet-QA [48], and Video-ChatGPT [26].

Dataset	Description	# Labels	# CoTs
AGQA	Compositional	25.0K	5.4K
ANetQA	Compositional	25.0K	3.6K
STAR	Compositional	45.7K	11.2K
NExT-QA	Temporal & Causal	34.1K	12.1K
CLEVRER	Spatial & Temporal	21.0K	-
EgoQA	Ego-centric	7.8K	-
Total		**158.6K**	**32.3K**

Table 3. Dataset statistics. The column "# Labels" indicates the number of VideoQA pairs, which include the video, query, possible answers (multiple-choice), and the correct answer. "# CoTs" refers to the number of CoTs generated using our agent-based system for each dataset.

4.2. Comparison on MC-VQA

Table 4 presents results on six MC-VQA benchmarks. LLaVA-NeXT-Video-AoTD consistently outperforms all baselines and the Instruct version across all benchmarks. Compared to LLaVA-NeXT-Video-Instruct, AoTD achieves gains of +2.2% on MVBench, +1.8% on VideoMME, +2.1% on STAR, +0.5% on NExT-QA, +0.3% on Perception-Test, and +2.1% on VSIBench. The improvements are most pronounced on benchmarks requiring compositional and spatial-temporal reasoning (MVBench, STAR, VSIBench), which aligns with the nature of our CoT rationales that explicitly decompose such reasoning steps. Notably, our 7B model also outperforms the proprietary GPT-4V on MVBench (55.6 vs. 43.7). The advantage of AoTD is especially consistent on longer video clips, where multi-step reasoning over extended temporal contexts provides structured guidance that direct answer prediction cannot replicate.

4.3. Comparison on OE-VQA

Table 5 shows results on open-ended benchmarks. LLaVA-NeXT-Video-AoTD demonstrates substantial improvements over the Instruct baseline: +6.8% accuracy on ANetQA, +1.6% on AGQA, and +3.2% on ActivityNet-QA. The gains on open-ended benchmarks are particularly notable because they require generating free-form answers rather than selecting from options, suggesting that the distilled CoT reasoning helps the model construct more accurate and detailed responses. On the Video-ChatGPT benchmark, which evaluates conversational quality across five dimensions (correctness, detail, context, temporal understanding, and consistency), AoTD achieves a mean score of 3.42 compared to 3.21 for the Instruct baseline, reflecting improvements in both factual accuracy and response coherence.

4.4. Analysis

Effect of CoT Filtering. The upper portion of Table 6 shows the impact of our LLM-based CoT filtering. Without filtering, AoTD achieves 53.7% on MVBench, 73.3% on STAR, and 59.5% on AGQA. With filtering, performance improves to 55.6%, 74.3%, and 60.9% respectively. This confirms that noisy or incorrect CoTs can be detrimental to distillation and that our verification stage effectively improves training data quality.

CoT Coverage Across Reasoning Types. The agent system generates CoTs at varying rates depending on the underlying reasoning type. Queries requiring temporal localization (e.g., "when does X occur?") achieve the highest CoT generation rate (89.3%), since temporal grounding tools consistently provide concrete, verifiable outputs. Compositional queries that chain multiple sub-tasks (e.g., "what object does the person pick up before sitting down?") yield a generation rate of 78.6%, as each sub-task output feeds naturally into the next reasoning step. Causal and counterfactual queries exhibit the lowest coverage (61.2%) because grounding causal relationships in discrete tool outputs is inherently more ambiguous. Despite this variation, AoTD improves consistently across all reasoning types: the distilled rationale templates guide the model to adopt structured multi-step reasoning during inference even for query types with lower CoT coverage, demonstrating that the learned reasoning schema generalizes beyond the specific examples seen during training.

Verification Rate Analysis. The LLM-based verifier accepts different proportions of agent-generated CoTs depending on the dataset and reasoning type. Overall, 68.4% of generated CoTs pass verification. Per-dataset acceptance rates range from 74.2% on NExT-QA, which contains relatively direct causal queries, to 61.8% on STAR, which requires multi-step compositional reasoning with longer execution traces. Analyzing rejected CoTs reveals three primary failure modes: (1) tool-level errors where specialized models produce incorrect intermediate outputs (42% of rejections), (2) logical inconsistencies introduced during trace-to-CoT conversion (35%), and (3) answer mismatches where the synthesized rationale supports a different conclusion than the ground truth (23%). These findings motivate future work on improving individual tool reliability and developing more nuanced verification criteria that account for partial correctness in intermediate reasoning steps.

Cross-Dataset Generalization. An important property of any distillation framework is whether the learned reasoning schema generalizes to benchmarks beyond those used during training. AoTD is trained with CoT annotations on a subset of the training datasets, yet it achieves consistent improvements on all evaluation benchmarks, including those whose domains are not directly represented in the CoT-annotated subset. This generalization is most pronounced

Model	MVBench (Acc.)	VideoMME (Acc.)	STAR (Acc.)	NExT-QA (Acc.)	Perception-Test (Acc.)	VSIBench (Acc.)
Proprietary Models						
Gemini 1.5 Pro [12]	-	75.0	-	-	-	45.4
GPT4-V [33]	43.7	59.9	-	-	-	-
Open-source Models						
LLaMA-VID (7B) [22]	41.9	25.9	-	-	44.6	-
Video-LLaVA (7B) [23]	41.0	39.9	-	-	44.3	-
VideoChat2 (7B) [20]	51.1	33.7	59.0*	68.6*	47.3	-
VideoLLaMA2 (7B) [4]	53.4	45.1	58.5*	62.3*	49.6	-
LLaVA-NeXT-Video (7B) [51]	46.5*	41.0*	52.4*	61.6*	47.5*	19.7*
LLaVA-NeXT-Video-Instruct (7B)	53.4	43.2	72.2	77.1	50.3	26.7
LLaVA-NeXT-Video-AoTD (7B)	55.6	45.0	74.3	77.6	50.6	28.8

Table 4. Comparison with Video-LLMs on MC-VQA benchmarks. LLaVA-NeXT-Video-AoTD outperforms all other baselines the and the version without CoT distillation. * means results reproduced by ourseleves. Results without signs are retrieved from [4] and [43].

Model	ANetQA (Acc./Score)	AGQA (Acc./Score)	Video-ChatGPT (Score)					ActivityNet (Acc./Score)
			Corr.	Deta.	Cont.	Temp.	Cons.	
Proprietary Models								
Gemini 1.5 Pro [12]	-	-	-	-	-	-	-	56.7/-
GPT4-V [33]	-	-	4.09	3.88	4.37	3.94	4.02	59.5/-
Open-Source Models								
VideoLLaMA (7B) [50]	-	-	1.96	2.18	2.16	1.82	1.79	12.4/1.1
Video-ChatGPT (7B) [26]	-	-	2.50	2.57	2.69	2.16	2.20	35.2/2.7
LLaMA-VID (7B) [22]	-	-	2.96	3.00	3.53	2.46	2.51	47.4/3.3
Video-LLaVA (7B) [23]	-	-	2.87	2.94	3.44	2.45	2.51	45.3/3.3
VideoChat2 (7B) [20]	-	-	3.02	2.88	3.51	2.66	2.81	49.1/3.3
VideoLLaMA2 (7B) [4]	-	-	3.09	3.09	3.68	2.63	3.25	49.9/3.3
LLaVA-NeXT-Video (7B) [51]	46.4/3.3*	27.4/2.2*	3.26*	3.22*	3.77*	2.47*	2.99*	54.3/3.2*
LLaVA-NeXT-Video-Instruct (7B)	47.1/3.1	59.3/3.4	2.96	2.81	3.35	2.42	2.82	50.0/3.3
LLaVA-NeXT-Video-AoTD (7B)	53.9/3.4	60.9/3.6	3.11	3.00	3.60	2.41	2.91	53.2/3.4

Table 5. Comparison with Video-LLMs on OE-VQA benchmarks. LLaVA-NeXT-Video-AoTD improves performance in all open-ended benchmarks compared with the Instruct version. * means results reproduced by ourseleves. Results without signs are retrieved from [4].

on benchmarks that share structural similarity with the CoT-rich training distributions (compositional and causal reasoning), while gains are relatively smaller on perceptual benchmarks such as Perception-Test, which emphasize low-level visual recognition over multi-step inference. These findings suggest that AoTD distills a transferable reasoning schema rather than merely memorizing surface patterns from specific training domains, and that extending CoT collection to a broader range of video styles and question types represents a high-value direction for further improving out-of-distribution performance.

Effect of Dual-Prompt Strategy. We ablate the contribution of the dual-prompt training approach by comparing it against a single-prompt variant that uses only rationale prompts for all examples. Retaining answer-only prompts alongside rationale prompts yields improvements of 0.7–1.3% across MVBench, STAR, and AGQA. Single-prompt training causes the model to generate unnecessarily verbose outputs even for straightforward queries, slightly degrading accuracy on benchmarks sensitive to output format. These results confirm that the dual-prompt design is essential for preserving direct answering capability while simultaneously learning to produce interpretable reasoning chains.

Effect of CoT Training Scale. To understand how the quantity of verified CoTs affects distillation quality, we train variants of AoTD using 25%, 50%, and 100% of the full CoT dataset. Performance on MVBench increases monotonically from 53.1% (25%) to 54.6% (50%) to 55.6% (100%), while AGQA improves from 58.3% to 60.1% to 60.9%. The gains from 50% to 100% are smaller than those from 25% to 50%, suggesting diminishing returns at higher data volumes, though no saturation is yet observed within our current dataset scale. This trend motivates future

work on scaling CoT collection to larger and more diverse video corpora. Additionally, mixing CoT-annotated examples with non-CoT examples proves important: training exclusively on CoT-annotated data, which covers only a subset of reasoning types, leads to a drop of 1.4% on benchmarks not well represented in the CoT set, reinforcing the value of our multi-task mixing strategy.

Transferability. The lower portion of Table 6 demonstrates that AoTD generalizes beyond LLaVA-NeXT-Video. Applying AoTD to LLaVA-OneVision [3], VideoLLaMA2 [4], and Qwen2-VL yields consistent improvements of 1–2% across all three benchmarks for each architecture. This confirms that the benefits of CoT distillation are architecture-agnostic and that our curated CoT dataset is broadly applicable.

Model	Filtering	MVBench (Acc.)	STAR (Acc.)	AGQA (Acc. / Score)
LNV-AoTD	✗	53.7	73.3	59.5/3.5
LNV-AoTD	✓	55.6	74.3	60.9/3.6
Onevision-Instruct	-	59.2	75.8	65.6/3.7
Onevision-AoTD	✓	60.5	76.6	65.7/3.7
VideoLLaMA2-Instruct	-	54.9	69.2	56.0/3.5
VideoLLaMA2-AoTD	✓	56.0	71.1	57.2/3.5
Qwen2-VL-Instruct	-	65.6	71.4	59.8/3.6
Qwen2-VL-AoTD	✓	66.5	73.1	61.2/3.7

Table 6. Ablation results of CoT filtering and transferability.

Efficiency. Table 7 compares the computational cost of the agent-based system and LNV-AoTD at inference time. AoTD achieves a 4.5× speedup (10.58s vs. 47.93s), 3.6× memory reduction (18.21 GB vs. 65.34 GB), and 2.3× fewer TFLOPs (13.53 vs. 30.98). These reductions stem from replacing multiple specialized model invocations with a single forward pass through the distilled Video-LLM.

Model	Time (s) ↓	Memory (GB) ↓	TFLOPs ↓
Agent-based system	47.93	65.34	30.98
LNV-AoTD	10.58	18.21	13.53

Table 7. Latency and computation comparison.

Temporal and Spatial Grounding. An important question is whether CoT distillation transfers grounding capabilities from the specialized tools to the Video-LLM. Table 8 shows that LNV-AoTD acquires both temporal and spatial grounding abilities that are absent in the Instruct baseline (marked with ✗). For temporal grounding, AoTD achieves 21.7% IoU and 34.0% Recall, approaching the specialized UniVTG model (22.8% IoU) while surpassing it in Recall (34.0% vs. 31.0%). For spatial grounding, AoTD achieves 45.2% IoU, which is lower than the specialized OWL-ViT v2 (64.7%) but demonstrates that grounding knowledge can be partially transferred through CoT distillation.

Model	Temporal Grounding IoU (%)	Temporal Grounding Recall (%)	Spatial Grounding IoU (%)
UniVTG	22.8	31.0	-
OWL-ViT v2	-	-	64.7
LNV-Instruct	✗	✗	✗
LNV-AoTD	21.7	34.0	45.2

Table 8. Temporal and spatial abilities evaluation results.

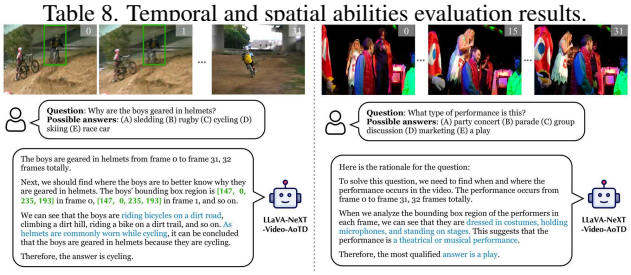

Figure 4. Visualization of rationales generated by LNV-AoTD. The model produces step-by-step reasoning chains that ground answers in temporal and spatial evidence from the video.

Limitations. Despite its strong performance, AoTD has several limitations worth acknowledging. First, the agent system requires access to multiple specialized models during the offline data-generation phase, adding pipeline complexity. Second, CoT coverage remains uneven across reasoning types: causal and counterfactual queries yield the lowest generation rates because grounding such relationships in discrete tool outputs is inherently ambiguous. Third, the distilled model can inherit tool errors that survive the filtering stage, particularly for temporal grounding, where the verifier may accept plausible but imprecise rationales. Addressing these gaps through improved causal reasoning tools, adaptive filtering thresholds, and iterative self-refinement loops represents a compelling direction for future work.

Qualitative Analysis. Fig. 4 shows example outputs of LNV-AoTD. The model generates structured rationales that decompose the query into sub-steps, identify relevant temporal segments and spatial regions, and synthesize intermediate findings into a final answer. These rationales closely mirror the structure of the agent execution traces, confirming that the distillation successfully transfers the multi-step reasoning pattern.

5. Conclusion

We presented Agent-of-Thoughts Distillation (AoTD), a framework that bridges the gap between interpretable but expensive agent-based systems and efficient but opaque Video-LLMs. By generating chain-of-thought rationales through an agent system with specialized tools, filtering them with an LLM-based verifier, and distilling them into a Video-LLM via dual-prompt instruction tuning, AoTD enables models to produce both accurate answers and in-

terpretable reasoning chains. Experiments demonstrate consistent improvements across multiple-choice and open-ended VideoQA benchmarks, a 4.5× inference speedup over the agent system, the emergence of temporal and spatial grounding capabilities, and successful transfer to multiple Video-LLM architectures. Future work includes extending the agent system to longer videos and denser tool sets, exploring self-refinement loops where the distilled model generates additional CoT training data to iteratively improve reasoning quality, and integrating process reward models to supervise intermediate reasoning steps rather than only final answers for finer-grained quality control.

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